

SHUDE CHEN

陳淑徳 (チンシュクトク)

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RESEARCH INTERESTS

Core areas: Machine Learning for Networked Systems; Multivariate Time-Series Forecasting; Anomaly Detection; Probabilistic Modelling and Gaussian Processes.

Future interests: Adaptive, robust, and computationally efficient learning for dynamic and distributed networked systems.

CURRENT POSITIONS

Systems Engineer, NSSOL Okinawa

Apr. 2026 – Present

NSSOL Business Division. Developing and administering enterprise workflows on ServiceNow, including AI-agent prototypes and AI-assisted workflow automation.

Visiting Researcher, University of the Ryukyus

Oct. 2025 – Present

Faculty of Engineering. Continuing research on network traffic forecasting and anomaly detection, with a focus on sparse Gaussian process models and multivariate traffic data.

EDUCATION

Master of Engineering, University of the Ryukyus

Apr. 2024 – Sep. 2025

Intelligent Information Program, Graduate School of Engineering and Science.

GPA: 4.0/4.0. Completed the two-year programme in 18 months.

Master's thesis: *Network Traffic Bandwidth Prediction and Anomaly Detection Using Multi-Output Gaussian Processes.*

多出力ガウス過程を用いたネットワークトラフィック帯域予測と異常検知.

Bachelor of Engineering, University of the Ryukyus

Apr. 2020 – Mar. 2024

Intelligent Information Course, Faculty of Engineering.

Completed the Global Shinryo Minor Program.

Bachelor's thesis: *Improving Accuracy of Network Bandwidth Prediction Using Multi-Channel Time-Series Data.*

多チャンネル時系列データを用いたネットワーク帯域予測の精度向上.

SELECTED RESEARCH EXPERIENCE

Network Traffic Forecasting and Anomaly Detection

Apr. 2023 – Present

University of the Ryukyus.

- Analysed real-world backbone network traffic data from the WIDE/MAWI Project.
- Developed multi-output Gaussian process models using inter-channel correlations and multi-scale feature augmentation, reducing prediction RMSE by up to 35.3% compared with single-output GPR in the evaluated MAWI experiments.
- Designed residual-based anomaly detection methods, achieving an F1-score of 0.930 on evaluated network traffic data.
- Investigated sparse Gaussian process approximations to reduce computational cost in multivariate network traffic analysis.

RESEARCH AND PROFESSIONAL EXPERIENCE

Teaching Assistant, University of the Ryukyus

Oct. 2024 – Sep. 2025

Supported undergraduate courses and student learning activities.

Technical Consultant, Information Technology Center

May 2024 – Sep. 2025

University of the Ryukyus. Provided technical support related to information systems and networks.

Research Intern, OIST

Aug. 2024 – Oct. 2024

Okinawa Institute of Science and Technology. Supported data processing for lower-limb exoskeleton rehabilitation robotics research.

Student Tutor, University of the Ryukyus

Jul. 2022 – Sep. 2025

Supported international students in their academic activities.

AWARDS

Best Presentation Award, IEEE ICICSP 2024

2024

Awarded for the presentation of research on multi-channel Gaussian process regression for network bandwidth prediction.

PUBLICATIONS

Journal Articles

- 1. [Enhanced Network Bandwidth Prediction with Multi-Output Gaussian Process Regression](#)
Shude Chen and Takayuki Nakachi
International Journal of Advanced Computer Science and Applications, 16(2), 2025
- 2. [Integrating Gaussian Process Regression and Correlated Signal Analysis for Anomaly Detection in Network Traffic](#)
Shude Chen and Takayuki Nakachi
International Journal of Computer Science and Network Security, 25(7), 17–29, 2025

Conference Papers

- 3. [Sparse Multi-output Gaussian Processes with Smoothed Residuals for Robust Anomaly Detection in Multivariate Network Traffic](#)
Shude Chen and Takayuki Nakachi
IEEE ICICSP 2025
- 4. [Multi-Channel Gaussian Process Regression for Network Traffic Anomaly Detection via Inter-Channel Correlation](#)
Shude Chen and Takayuki Nakachi
ITC-CSCC 2025
- 5. [Improving Accuracy of Network Bandwidth Prediction Using Multi-Channel Gaussian Process Regression](#)
Shude Chen and Takayuki Nakachi
IEEE ICICSP 2024 – Best Presentation Award

Technical Reports (in Japanese)

- 6. [多出力ガウス過程を用いたネットワーク帯域予測の精度向上](#)
陳淑德, 城間政司, 名嘉村盛和, 仲地孝之
IEICE Technical Report, SIS2024-39, 124(288), 49–54, 2024 (in Japanese)

TECHNICAL SKILLS

Programming	Python; JavaScript; C/C++; R; HTML/CSS
Machine Learning & Data Analysis	Gaussian Process Regression; Time-Series Forecasting; Anomaly Detection; Multivariate Data Analysis; Network Traffic Analysis; Model Evaluation
Libraries & Frameworks	PyTorch; GPyTorch; scikit-learn; NumPy; pandas; Matplotlib
Platforms & Tools	ServiceNow; Git/GitHub; Jupyter Notebook; Google Colab; LaTeX
Parallel Computing	CUDA (Basic); SYCL (Basic)
Enterprise Systems	SAP (Basic)

LANGUAGES

Chinese	Native
Japanese	Professional Working Proficiency (JLPT N1, 2024)
English	TOEIC Listening & Reading: 720